

Module 4: Supervised Learning
Submodule: 4.6 Gaussian Discriminant Analysis
Lecture: Discriminative vs Generative Learning
Algorithms

XCS229 Machine Learning

[Tengu Ma]

Contents

I	Introduction	1
II	Review of Gaussian Discriminant Analysis	1
III	Parameter Estimation	2
IV	Comparison with Logistic Regression	3
V	Summary	5

I. Introduction

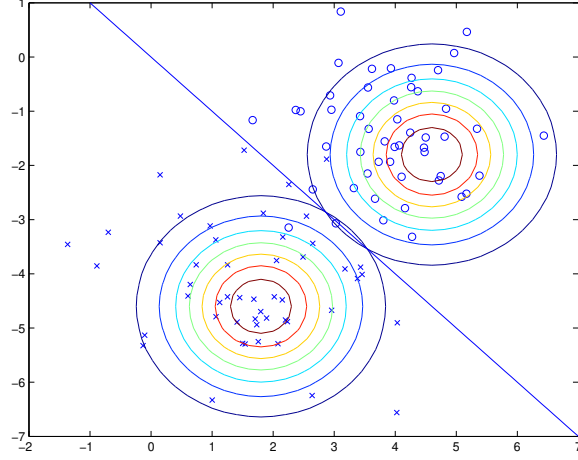
In this lecture, we discuss the Gaussian Discriminant Analysis (GDA) reviewed from our previous session and explore a new case concerning spam filtering, focusing on a discrete, rather than continuous, random variable x .

II. Review of Gaussian Discriminant Analysis

In GDA, we model the conditional probability $p(x|y)$ and the prior probability $p(y)$. Under the framework of GDA:

$$\begin{aligned}x \mid (y = 0) &\sim \mathcal{N}(\mu_0, \Sigma). \\x \mid (y = 1) &\sim \mathcal{N}(\mu_1, \Sigma).\end{aligned}$$

Also, $p(y = 1) = \phi$. With graphical representation could depict the clustering of positive and negative samples.



III. Parameter Estimation

To estimate the parameters ϕ , μ_0 , μ_1 and Σ , we employ the Maximum Likelihood Estimation (MLE) method. The objective is to maximize the log-likelihood function:

$$\arg \max_{\phi, \mu_0, \mu_1, \Sigma} \sum_{i=1}^n \log p \left(x^{(i)} | y^{(i)}; \mu_0, \mu_1, \Sigma \right) + \log p \left(y^{(i)}; \phi \right)$$

The derived parameters from the MLE include:

- Probability of positive class:

$$\phi = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}$$

- Means:

$$\mu_0 = \frac{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 0\} \cdot x^{(i)}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 0\}}$$

$$\mu_1 = \frac{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\} \cdot x^{(i)}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}}$$

- Covariance:

$$\Sigma = \frac{1}{n} \left(\sum_{i \in U_0} \left(x^{(i)} - \mu_0 \right) \left(x^{(i)} - \mu_0 \right)^T + \sum_{i \in U_1} \left(x^{(i)} - \mu_1 \right) \left(x^{(i)} - \mu_1 \right)^T \right)$$

After learning the parameters, we make predictions for a new example x . The task is to compute the most likely value of y given x :

$$\arg \max_{y \in \{0,1\}} p(y|x; \phi, \mu_0, \mu_1, \Sigma)$$

The decision boundary is characterized by the equality:

$$\{x : p(y = 1|x) = p(y = 0|x) = 0.5\}$$

This results in the linear decision boundary:

$$\{x : \theta^T x + \theta_0 = 0\}$$

where θ and θ_0 are derived as functions of ϕ, μ_0, μ_1 and Σ .

IV. Comparison with Logistic Regression

The primary distinctions between GDA and logistic regression lie in:

- GDA makes strong assumptions about the underlying distributions of $p(x|y)$ (Gaussian).

$$\begin{aligned} x \mid (y = 0) &\sim \mathcal{N}(\mu_0, \Sigma). \\ x \mid (y = 1) &\sim \mathcal{N}(\mu_1, \Sigma). \\ y &\sim \text{Bernoulli}(\phi). \end{aligned}$$

- Logistic regression directly models $p(y|x)$ without assuming distributions for x . In logistic regression, the model takes the form:

$$p(y = 1|x) = \frac{1}{1 + e^{-(\theta^T x + \theta_0)}}$$

Thus, the differences in model assumptions lead to potential variances in the derived parameters θ and θ_0 .

When determining which model to prefer between Gaussian Discriminant Analysis (GDA) and Logistic Regression, several factors come into play:

- **Decision Boundaries and Model Assumptions:** GDA and logistic regression will, in general, produce different decision boundaries when trained on the same dataset. It is important to consider:
 - If the conditional distribution $p(x|y)$ is multivariate Gaussian (with shared covariance Σ), then $p(y|x)$ necessarily follows a logistic function.

- Conversely, $p(y|x)$ being a logistic function does not imply that $p(x|y)$ is multivariate Gaussian. This demonstrates that GDA makes stronger modeling assumptions about the data than logistic regression.
- **Asymptotic Efficiency of GDA:** GDA has certain advantages under specific conditions:
 - When the modeling assumptions of GDA are correct, particularly when $p(x|y)$ is indeed Gaussian (with shared Σ), GDA is considered **asymptotically efficient**.
 - This means that with large training sets ($n \rightarrow \infty$), GDA will outperform any other algorithm in accurately estimating $p(y|x)$.
 - GDA is generally expected to provide better performance even with smaller training set sizes if the Gaussian assumption holds.
- **Robustness of Logistic Regression:** On the other hand, logistic regression has its unique strengths:
 - Logistic regression makes significantly weaker assumptions about the data distributions, making it more robust to incorrect modeling assumptions.
 - There exist various distributions for $p(x|y)$ that could result in $p(y|x)$ taking the form of a logistic function. For example, if, $x \in \mathbb{N}$ (i.e., x is an integer):

$$\begin{aligned} x|y = 0 &\sim \text{Poisson}(\lambda_0), \\ x|y = 1 &\sim \text{Poisson}(\lambda_1), \end{aligned}$$

then $p(y|x)$ will still be logistic.

In summary:

- GDA:
 - * Stronger modeling assumptions.
 - * More data efficient: requires less training data to achieve good performance when assumptions are correct.
- Logistic Regression:
 - * Weaker assumptions, making it significantly more robust to modeling deviations.
 - * Generally performs better than GDA when the data distribution is non-Gaussian, especially as dataset size increases.

Due to its stronger robustness and lower sensitivity to violations of the underlying assumptions, logistic regression is often preferred in practice over GDA.

V. Summary

- GDA provides a framework for modeling when the underlying assumptions (Gaussian distributions) are valid, and can yield better estimates with sufficient data.
- The decision boundary derived from GDA can be expressed linearly, reflecting its nature as a generative model.
- Logistic regression offers greater flexibility and robustness, making fewer assumptions about the data distribution and often leading to better performance in varied scenarios.
- The choice between GDA and logistic regression should consider the nature of the data, the validity of assumptions about the underlying distributions, and the amount of available data.